IGPT: Imitation Game Personal Texter

Stanford CS229 Project

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1 Introduction

In today's fast-paced world, keeping up with text messages from friends and family can be challenging. Many of us are often too busy to respond promptly, leading to delays in communication. This can unintentionally make loved ones feel neglected or unappreciated. To address this problem, we create an automated text response system capable of mimicking an individual's unique communication style.

Our project focuses on one-to-one conversations, excluding group chats. Using this data, we aim to generate contextually appropriate responses that emulate the user's texting style. To achieve this, we will explore various approaches, including fine-tuning pre-trained large language models and employing in-context learning with state-of-the-art LLMs. We will use a Naive Bayes model, human evaluation, and correlation evaluation to test our models.

2 Related Work

The development of an automated text-reply system capable of mimicking an individual's communication style draws from key advancements in natural language processing and language modeling. Central to this progress is the Transformer architecture, introduced by Vaswani (2017) in their seminal paper. This paper revolutionized NLP by introducing a self-attention mechanism that allows models to focus on relevant parts of an input sequence, enabling state-of-the-art performance in tasks such as text generation and style adaptation. This foundation underpins most modern large language models.

Building on these advances, Chen and Moscholios (2024) explore how prompting techniques can tailor responses from LLMs to replicate an individual's unique communication patterns. This work aligns closely with our goal of generating personalized responses by leveraging in-context learning to fine-tune the conversational tone and style of an LLM without extensive modifications. It provides crucial insights into prompt engineering, which we incorporate in our system to achieve stylistic fidelity.

Finally, Ren et al. (2024) emphasize the importance of aligning model responses with a user's preferred communication style during fine-tuning. This approach not only enhances user satisfaction but also ensures more natural and engaging interactions. The techniques discussed, including style-specific loss adjustments and post-training fine-tuning, inspired our method for refining LLM outputs to maintain consistency with individual texting habits.

Together, these works provide the theoretical and practical framework for designing a system that leverages cutting-edge NLP techniques to address the challenges of personalized automated communication.

3 Dataset and Features

Our dataset was constructed using text messages collected from each team member's preferred communication platforms.

Bruno Dumont (WhatsApp): Data was exported using WhatsApp's "Export Chat" feature, which generates a .txt file containing all messages with timestamps and sender information. After export, we processed the data into a back-and-forth conversation structure. We group consecutive messages from the same sender into a separate .txt file containing only the text content without sender metadata.

Emily Xia/Eric Chen (Discord): Data was obtained using Discord's "Request Data" feature. The exported data contains only the sender's messages over all time, which were then processed into .txt files and tokenized for compatibility with language models.

Data cleaning included removing extraneous metadata, formatting inconsistencies, and irrelevant symbols. For example, some of our discord messages consisted of only bot commands (e.g. "pls fish catch"). We removed these commands because they do not provide any information on the linguistic patterns of our text. In total, we had 252175 datapoints for Emily, 131009 datapoints for Eric, and 11426 for Bruno.

4 Generation Methods

4.1 Fine Tuning

For our Project Milestone, we chose to fine-tune GPT-2 as it was small enough to train with limited compute. However, we quickly noticed that due to the limited size of GPT-2, the outputs were often simply non-sensical. For our final project, we successfully fine-tuned the larger, 7-billion parameter Llama-2-7b Chat, a fine tuned version of Llama-2 that is optimized for dialogue.

We leveraged LoRA (Low-Rank Adaptation of Large Language Models, (Hu et al., 2021)) to successfully train Llama-2 with limited compute. LoRA freezes the pretrained model weights and injects trainable matrices into layers of the architecture in order reduce the number of training parameters. For the `target_modules` of LoRA, we chose `q_proj` and `v_proj`. The `q_proj` module is used when the attention mechanism's query vectors need adaptation. This is helpful since we want to adapt this model to our own texting language and style. `v_proj` is often used for semantic analysis which we used since we want the outputted text messages to match our general tone and personalities.

We chose solely to fine-tune a model for Emily, as she had the largest corpus of data. We used a batch size of 4 and only trained with 3 epochs. We used 20% of the dataset as an eval set. Given that we had a suite of final evaluation methods, we did not create a test set.

4.2 Prompting

An alternative method we employed to obtain models personalized to our texting styles was in-context learning. In-context learning (ICL) allows LLMs to generate outputs tailored to a task by conditioning on specific instructions or examples provided in the prompt, as shown in Figure 1 (Brown, 2020). We employed a form of ICL, wherein we included the entire dataset of a human's messages along with instructions to emulate their texting style.

Through ICL, we provided the model a detailed representation of the a person's linguistic patterns, such as sentence structure, word choice, and tone. We found that this was often more convincing and reliable than the fine-tuned models. We suspect that this is due to two main reasons: ChatGPT-4o is significantly larger than Llama-2 7B Chat, so it is better able to adapt itself to situations like this; additionally, LLMs are generally more influenced by items placed in their immediate context (i.e. the prompt) than in their pretraining Brown (2020).

An additional benefit of in-context learning is that we can personalize our models without going through an expensive fine-tuning process. We display our results from ICL as well as a comparison with fine-tuning in in Section 6.

Forget every other conversation to do this task. Pretend to be Eric Chen. You will have access to all his messages, and you will use them to learn his texting style and emulate it. Then, answer the following questions like Eric Chen would text a friend.

The previous messages are attached as a jsonl file. Answer the questions below:

Questions: What's been on your mind lately? ...

Figure 1: Prompt used in ICL

4.3 Naive Bayes Algorithm

The Naive Bayes model is a simple probabilistic classifier that predicts the most likely source of a message by assuming each word in the text contributes independently to the overall probability of the sender. In our project, we employ 3 separate Naive Bayes models to distinguish between messages written by a specific human and those from other people. Each model is trained on a dataset with text messages labeled with the individual's name or "other".

We processed the messages to calculate word frequencies separately for the user's texts and the model's, creating a probability distribution for each group. To handle unseen words and ensure smooth predictions, we applied Laplace Smoothing. After training, we calculated the probability of statements based on their component words. We then use maximum likelihood to predict which class the message is from. This model played a crucial role in our evaluation process, offering an additional metric for assessing the similarity of mimicked and real messages.

5 Evaluation Methods

To evaluate the performance of our in-context learning and fine-tuned models, we designed experiments that assess how effectively the models replicate the user's unique texting style. Directly comparing the outputs to human outputs was largely impractical due to the large variability of human language and a lack of textual context. Instead, we implemented three evaluation methods: human evaluation, Naive Bayes evaluation, and correlation analysis. For consistency, we standardized the testing by creating 25 questions representative of typical texting scenarios, such as as "Do you feel like going out later?" or "What's your favorite thing to do on a lazy day." Each model and the user responded to the same set of questions, ensuring comparability across all evaluation methods.

5.1 Human Evaluation

We reached out to a few of our close friends who are familiar with our texting styles and asked them to attempt to discriminate between the model-generated and human-generated texts. Using a Google Form (Emily's form, Eric's form), evaluators were presented with each question alongside one randomly selected answer—either from the user, the fine-tuned model, or the in-context learning model. They were informed that some responses were generated by the user and others by models, and their task was to guess who wrote each response. By relying on people familiar with the user's texting habits, this evaluation provided qualitative insights into how convincingly each model replicated the user's style.

5.2 Naive Bayes Evaluation:

We employed the Naive Bayes model, described in Section 4.3, to measure how successful our models are at mimicking our texting styles. Using a maximum likelihood classification, we collected the Naive Bayes algorithm's predictions across both human and model-generated texts and recorded its accuracies.

Question: How are you doing?

Emily: "im doing p good, wbu?"

Bruno: "I'm good just chilling tbh. How about u?"

Eric: "i'm pretty good lol wbu"

Llama2 7B Fine Tune: "i'm doing well"

GPT 4o ICL: "pretty chill, hbu?"

Figure 2: Multiple human and model responses to "How are you doing?"

5.3 Correlation Evaluation:

To gain further insight into language similarity between models and humans, we analyzed the embedding differences between user responses and model outputs to the same questions. For a baseline comparison, we also calculated the difference between the original human text and the response of two other texters.

We chose questions for which the user and model responses had relatively similar semantic meaning. For example, the prompt "How are you doing?" all resulted in some variation of "good" for the original user and the models as shown in Figure 2. With overarching semantic differences aside, the embedding differences can focus on word usage and syntax, thus capturing differences between our true texting style and that of the models. Using Bidirectional Encoder Representations from Transformers (BERT), we tokenized each string and extracted embeddings from the last hidden state for individual token embeddings. We then found the cosine similarity between the origin human response and model/baseline responses. The cosine similarity is a value between -1 and 1 with 1 being perfectly similar, 0 being no similarity, and -1 being perfectly dissimilar. These evaluations together offered both subjective and objective insights, allowing us to comprehensively compare the performance of our in-context learning and fine-tuned models.

6 Experiments / Results / Discussion

6.1 Human Evaluators vs Naive Bayes

The results Emily's and Eric's Google Form surveys are in Figures 4, and 5, respectively. The average accuracy for Emily's and Eric's experiments were 61.17% and 57.28%, respectively—barely better than random guessing. This indicates that our model outputs successfully matched our tone and language enough to fool even our closest friends. Our Naive Bayes model obtained a 64% accuracy on classifying both Emily's and Eric's results.

For both ICL GPT 4o and fine-tuned Llama2 7B, we see that Naive Bayes performed better at classifying messages than human evaluators in nearly all categories. However, it notably receives a lower accuracy for Emily's human-generated texts. One possible explanation for this discrepancy is that Emily's data spans six years while Eric's data is mostly from the past two. As a result, the model is learning from an outdated version of Emily, creating the difference.

6.2 Correlation Evaluation

As shown in Figure 1, both ICL and our fine tuned model had greater similarities to the original texter than the two other individuals, showing success in our training efforts. Of the two methods, ICL resulted in a higher correlation value.

Analyzing the results from our tests, we observed that each evaluation method emphasized distinct aspects of the generated responses. The Naive Bayes model prioritized word choice and syntax without considering the broader semantic meaning of the sentences, while the correlation method focused solely on semantic alignment. Human evaluation combined multiple factors, leveraging intuition and subjective interpretation. Notably, the Naive Bayes model generally outperformed

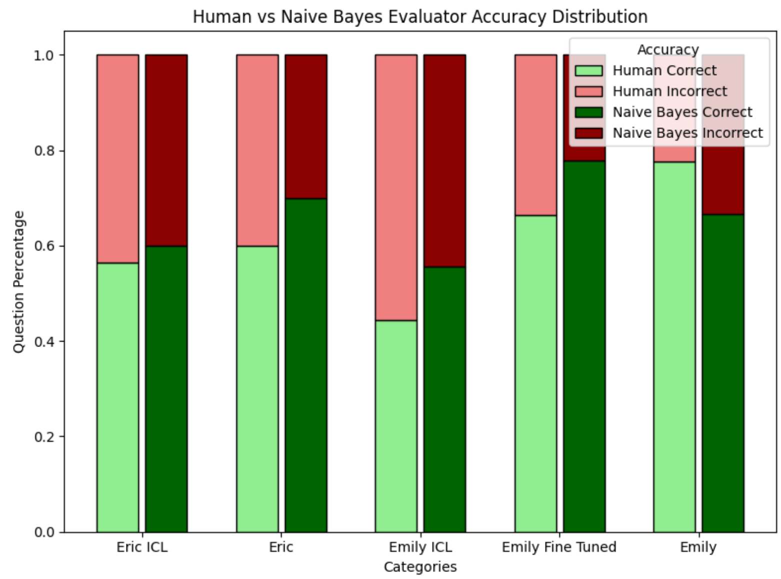

Figure 3: Human Evaluator and Naive Bayes results (values are accuracies for ease of comparison). These scores represent the proportion of model-generated ("ICL" or "Fine Tuned") and the proportion of human-generated ("Eric" or "Emily") responses that were correctly classified.

Prompt	**Fine Tuned**	**ICL**	**Bruno**	**Eric**
How are you doing?	0.827	0.868	0.885	0.824
Do you want to join me for dinner?	0.773	0.892	0.719	0.770
Do you feel like going for a walk?	0.859	0.802	0.832	0.879
What's your favorite thing to do on a lazy day?	0.832	0.867	0.810	0.708
Average	0.823	0.857	0.812	0.795

Table 1: Cosine similarities of various models on

human evaluators. This finding suggests that linguistic patterns, independent of semantic context, can provide valuable insights into verifying the authenticity of text.

7 Conclusion / Future Work

In this paper, we evaluated two approaches—In-Context Learning (ICL) and fine-tuning—to generate user-specific text responses. ICL GPT 4o emerges as the highest-performing method. The computational advantage that GPT 4o gains by possessing over 200B parameters (compared to Llama's 7B), along with the contextual advantage that ICL has over fine-tuning, results in a significantly more effective model. Though fine-tuning performed well, it was slightly less consistent as it was constrained by the limitations of pre-trained weights.

For future work, fine-tuning larger, more advanced language models could further close the performance gap. Additionally, we could expand our dataset with other sources (e.g. iMessage and Instagram DMs) would enhance the model's versatility across different social contexts. Third, incorporating multimodal elements like images, emojis, and formatting styles could significantly improve response authenticity. Finally, implementing advanced learning methods and evaluation frameworks such as reinforcement learning with human feedback could optimize model performance for specific use cases.

The future of human-AI interaction beckons, not as a replacement for human discourse, but as a bridge to enhance it.

8 Appendices

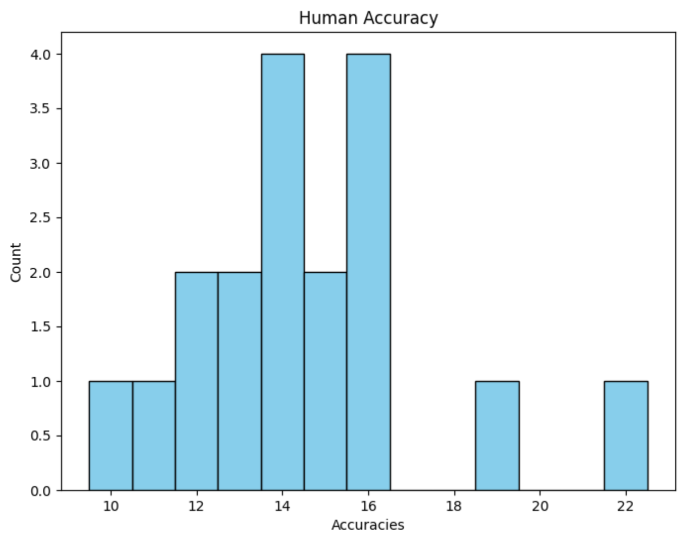

Figure 4: Emily's Human Evaluator Accuracy (total of 25)

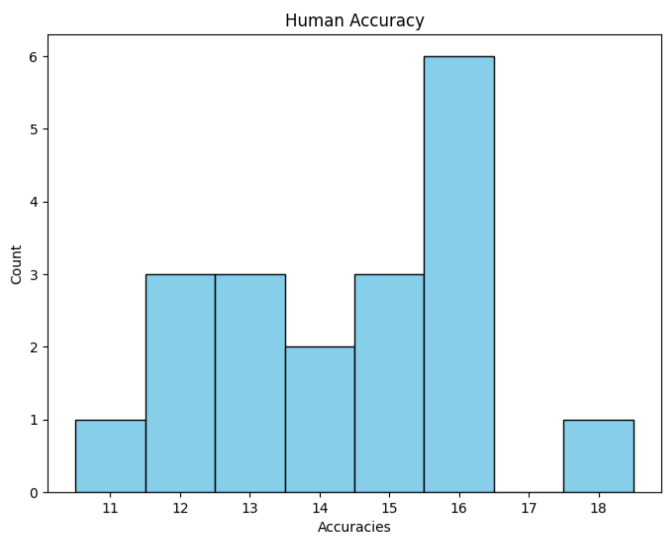

Figure 5: Eric's Human Evaluator Accuracy (total of 25)

9 Contributions

All team members contributed equally to this project through frequently communication and feedback.

Eric mainly worked on the GPT-4o ICL and fine-tuning parts of the experiments. He solicited compute resources for the group and wrote parts of and thoroughly edited the milestone submission and the final paper.

Bruno worked in GPT-4o ICL, developing getting the data and running the Naive Bayes Model, designing, formalizing and running the experiments, gathering parts of the data as well as substantial writing in both the milestone and final report.

Emily worked on processing and cleaning data, navigating Google Cloud Storage for storing the trained models, fine tuning Llama, and analyzing + visualizing the human evaluation results. Additionally, she also worked on writing and editing the final paper.

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